# SWAHIVO LOGISTICS OS

## MASTER UI RECONSTRUCTION & FULL-STACK IMPLEMENTATION PROMPT

### Admin + Customer + Warehouse + Driver Dashboards and All Child Pages

**PROJECT TYPE: SWAHIVO LOGISTICS**
**DESIGN AUTHORITY:** All 18 Uploaded Reference Screenshots
**IMPLEMENTATION STANDARD:** High-Fidelity Screenshot Reconstruction
**BACKEND:** Convex — Real Database and Business Operations
**UI RESTRICTION:** No Popup-Card-Based Workflows
**BRAND:** SWAHIVO Logistics OS

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# 1. CRITICAL INSTRUCTIONS — READ FIRST

You are acting as a senior enterprise SaaS frontend architect, UI reconstruction engineer, logistics systems engineer, Convex backend engineer, and quality assurance specialist.

I have uploaded screenshots showing exactly how my logistics SaaS application must look.

**YOUR TASK IS TO RECONSTRUCT THESE DESIGNS, NOT TO REDESIGN THEM.**

The attached screenshots are the visual authority for this project.

You are not being asked to suggest a new interface, generate an alternative design, or develop a simplified prototype.

You must implement the layouts, component structures, page compositions, sidebar navigation, tables, charts, maps, forms, tabs, operational panels, and visual hierarchy shown in the uploaded images.

Where the screenshot shows a particular arrangement, preserve it.

Where the screenshot provides no information, derive the missing interface from the nearest applicable uploaded reference, using the same design system and structural conventions.

Never substitute your personal UI preferences for the provided designs.

## NON-NEGOTIABLE RULES

1. Do not create a new application.
2. Do not rebuild my existing project.
3. Do not replace the technology stack.
4. Do not redesign the uploaded screenshots.
5. Do not create generic dashboard templates.
6. Do not replace full operational pages with popup cards.
7. Do not hide entire workflows inside modal dialogs.
8. Do not create decorative, nonfunctional navigation.
9. Do not put the same dataset on every status tab.
10. Do not use fake authentication or simulated backend success.
11. Do not use mock operational data in production.
12. Do not generate fake GPS locations or QR codes.
13. Do not omit any major section visible in a reference image.
14. Do not copy placeholder statistics from reference images as live data.
15. Do not modify working application modules unnecessarily.
16. Do not declare a page complete before verifying its appearance and functionality.

**You are required to follow the references even when your default design preferences differ.**

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# 2. KEEP MY EXISTING TECHNOLOGY STACK

The project already uses:

- Vite
- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Convex backend and database

Preserve the existing:

- Project directory structure where practical
- React Router configuration or current routing architecture

- Authentication integration
- Convex schema and functions
- Organization and tenant model
- Existing working components
- Shipment and warehouse business logic
- Real-time synchronization
- Shipping documents
- Backend integrations
- Deployment configuration

Do not introduce Next.js, Supabase, Firebase or another primary database.

Suggested libraries, only where needed and compatible with the current project:

| Feature | Implementation |
|---|---|
| Charts | Existing chart library, or Recharts |
| Icons | Lucide React |
| Warehouse 2D Map | react-konva + konva |
| QR Generator | qrcode |
| QR Scanner | @zxing/browser |
| Printing | react-to-print |
| Real geographic maps | Existing map library; otherwise a suitable MapLibre or Leaflet integration |
| Server data | Convex queries and mutations |
| Real-time changes | Convex reactive subscriptions |
| Testing | Vitest + Playwright |

Do not install redundant packages when equivalent working libraries already exist.

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# 3. SCREENSHOT INVENTORY — MANDATORY REFERENCE MAPPING

You will receive 18 reference images.

Inspect every image at its original resolution.

Before writing UI code, identify:

- Main layout grid
- Left sidebar structure
- Top navigation
- Page heading
- Breadcrumbs
- KPI card positions
- Content panel arrangement
- Tables and columns
- Tabs
- Action buttons
- Search and filter controls
- Maps
- Charts
- Detail panels
- Status colors
- Forms
- Pagination
- Responsive implications

# Primary Overview References

**REF-01 — Admin Master Reference**

`TZLogistics Admin Dashboard Overview(2).png`

Use as the authoritative reference for the application's master dashboard styling and Admin overview.

**REF-02 — Customer Dashboard**

`SWAHIVO Logistics Customer Dashboard(1).png`

Use for customer-specific content, page composition and navigation.

**REF-03 — Additional Customer Operations Reference**

`TZLogistics Operations Dashboard(2).png`

Use for detailed Customer navigation, warehouse inventory, storage bookings, shipments, payments, tracking and activity widgets.

**REF-04 — Warehouse Manager Dashboard**

`TZLogistics Warehouse Operations Dashboard(2).png`

Use for warehouse operations overview and its primary menu.

**REF-05 — Driver Dashboard**

`TZLogistics Driver Dashboard(2).png`

Use for driver-specific navigation, assigned shipments, trip routes, pickup operations and delivery workflows.

**REF-06 — Public Landing Page**

`TZLogistics Warehouse & Shipping Platform(2).png`

This is a separate public-facing website design, not a dashboard.

# Detailed Child-Page References

**REF-07 — Shipments**

`SWAHIVO Logistics Shipment Dashboard(1).png`

**REF-08 — Create Shipment**

`SWAHIVO Shipment Creation Dashboard(1).png`

**REF-09 — Customers / CRM**

`Customers Dashboard image: SWAHIVO Logistics Customer Dashboard(1).png` is the customer overview reference; also use the separately supplied Customers CRM screenshot if available in the upload set.

**REF-10 — Transporter Management**

`Transporter Management Dashboard(1).png`

**REF-11 — Billing**

`SWAHIVO Billing Dashboard Overview(1).png`

**REF-12 — Warehouse Reports**

`Swahivo Logistics Reports Dashboard(1).png`

**REF-13 — Warehouse Map Designer**

`Warehouse Map Designer Dashboard(1).png`

**REF-14 — Racks & Bins**

`Warehouse Racks & Bins Dashboard(1).png`

**REF-15 — Receiving & Putaway**

`SWAHIVO Warehouse Receiving Dashboard(1).png`

**REF-16 — QR Labels & Scanner**

`Warehouse QR Labels Dashboard(1).png`

**REF-17 — Package Locator**

`SWAHIVO Warehouse Package Locator Dashboard(1).png`

**REF-18 — Transfers & Picking**

`Warehouse Transfers & Picking Dashboard(1).png`

**REF-19 — Cycle Counts**

`Warehouse Cycle Count Dashboard(1).png`

Note: REF numbers represent design reference categories. Some references serve more than one screen. Inspect the actual 18 attached files rather than assuming there are 19 separate images.

If an explicitly referenced screenshot is not accessible, report its filename and use the closest available supplied design until the missing reference is available. Never claim to have inspected an inaccessible file.

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# 4. MASTER DESIGN SYSTEM — ADMIN DASHBOARD

The file:

`TZLogistics Admin Dashboard Overview(2).png`

defines the shared visual conventions.

## Desktop Dashboard Structure

Follow this arrangement:

1. Fixed dark navy sidebar on the left.
2. Full-width top navigation across the content area.
3. Main page heading and subtitle.
4. Action controls aligned to the upper right.
5. A single horizontal row of summary KPI cards where space permits.
6. A structured dashboard content grid.
7. Charts, maps and tables organized exactly like the reference.
8. Compact secondary panels.
9. Consistent page padding and alignment.

At approximately 1536 × 1024 resolution, reproduce the reference's desktop composition.

### Sidebar

Use a dark navy background.

Maintain:

- White navigation text
- Thin outlined navigation icons
- Active item with strong blue highlight
- Clear section headings
- Compact row spacing
- Nested expandable navigation
- Consistent hover and active states
- Appropriate scroll behavior

Use an approximately 220–235px sidebar on desktop, adjusted only when necessary to match the reference.

Do not use a floating sidebar.

Do not replace the sidebar with a horizontal navigation menu on desktop.

## Header

Recreate the reference header structure:

- Global search
- Notification icon and unread indicator
- Messaging shortcut when applicable
- Language selection where configured
- Current authenticated user avatar
- Current user's role
- Profile controls

## Typography

Maintain a clean, professional sans-serif UI.

Reuse the existing font configuration wherever possible.

Use:

- Bold dark page titles
- Medium-weight card headings
- Compact table typography
- Muted metadata
- Clear numerical emphasis
- Restrained color contrast

Do not introduce decorative typography, oversized headings or excessive whitespace.

## Content Styling

Match:

- Light gray-blue main background
- White rectangular content cards
- Subtle card boundaries
- Small, consistent corner radii
- Moderate internal padding
- Blue primary action buttons
- Compact search fields
- Small status badges

- Professional data tables

Do not redesign this into a glassmorphism, neomorphism or oversized-card layout.

### KPI Cards

Maintain the screenshot's format:

Icon → Label → Large Numerical Value → Comparison / Supporting Metric

Use blue, green, purple, orange, red and yellow only where the reference indicates meaningful information categories.

KPI cards remain permanently positioned within the page layout.

They are NOT popup cards.

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# 5. STRICT NO-POPUP-CARD POLICY

This requirement applies across all dashboards.

**Do not create entire pages inside floating popup cards, modal dialogs, overlay windows or temporary drawers.**

Examples of prohibited approaches:

- Clicking Create Shipment opens a large modal containing the full form.
- Clicking Warehouse Map opens a popup card instead of the real map page.
- Clicking Customers opens a floating detail card instead of the customer management page.
- Clicking Reports opens a modal report viewer.
- Clicking Racks & Bins opens a floating overlay instead of the reference's structured screen.
- Clicking Billing opens a payment popup replacing the billing page.

Use these patterns instead:

**Primary actions:** Navigate to dedicated full pages.

**Record details:** Use routed detail pages or permanent inline detail panels, following the corresponding screenshot.

**Multi-step forms:** Use full-page forms with visible progress indicators.

**Data filtering:** Use in-page controls.

**Editing:** Use dedicated edit routes or inline editors.

**Confirmations:** Use inline confirmation sections or clearly designated confirmation pages for significant operations.

**Status feedback:** Use inline status messages and validation feedback.

Compact dropdowns required by filters or navigation are acceptable.

KPI cards and permanent information panels visible in the screenshots must remain.

The phrase "no popup cards" must not be misinterpreted as "remove all dashboard cards."

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# 6. ADMIN DASHBOARD — IMPLEMENT EXACT REFERENCE STRUCTURE

Use `TZLogistics Admin Dashboard Overview(2).png`.

## Admin Overview

Recreate the top row:

- Total Shipments
- Active Storage
- In Transit
- Delivered
- Pending Actions
- Revenue

Next reproduce the content structure:

**Upper section**

- Shipments Overview chart
- Warehouse Capacity chart and utilization breakdown
- Live Tracking map

**Middle section**

- Storage Bookings summary
- Recent Invoices
- Recent Activities

**Lower section**

- Recent Shipments table

Follow the relative proportions, chart placement, tables and visual hierarchy shown in the screenshot.

All numbers must come from actual Convex data.

# Admin Sidebar

Preserve this navigation structure:

## Main

- Dashboard

## Operations

- Shipments
- Create Shipment
- Pickup Management
- Delivery Management
- Warehouse Management
    - Warehouses
    - Storage Bookings
    - Receiving
    - Inventory
    - Warehouse → Shipping
    - Stock Movements
    - Storage Rates
    - Locations & Bins
- Customers
- Drivers & Vehicles
- Routes & Coverage

Integrate Transporter Management into the existing carrier/provider navigation without creating unnecessary duplicate modules.

## Finance

- Quotes & Pricing
- Invoices & Payments
- Reports & Analytics

### System

- Users & Permissions
- Notifications
- Audit Logs
- System Settings

Do not remove working navigation items already present in the application.

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# 7. CUSTOMER DASHBOARD — COMPLETE RECONSTRUCTION

Use:

`SWAHIVO Logistics Customer Dashboard(1).png`

and:

`TZLogistics Operations Dashboard(2).png`

These screenshots define the Customer dashboard's business content and structure.

Recreate the visual styling using the shared Admin design system, while retaining the Customer dashboard's layout, menu and capabilities.

## Customer Overview

Top section:

- Welcome heading
- Warehouse Storage Booking button
- Create Shipment button

KPI row:

- Active Storage

- Total Bookings
- Active Shipments
- Delivered
- Pending Payments
- Total Spent

Main workspace:

- My Warehouse Inventory
- Storage Bookings
- My Shipments
- Track Your Shipment
- Recent Invoices
- Recent Activity

Follow the screenshots' actual table columns, content groupings and card arrangement.

# Customer Sidebar

- Overview
- Warehouse Bookings
- My Inventory
- Request Dispatch
- My Shipments
- Create Shipment
- Track Shipment
- Payments & Invoices
- Notifications
- Support
- Settings

# Implement Customer Child Pages

## Warehouse Bookings

Full-page booking management.

Provide:

- List of bookings
- Active/Pending/Completed status filters
- Booking details
- New storage booking
- Warehouse selection

- Cargo and quantities
- Dates
- Storage estimate
- Booking submission
- Booking history

Link the booking to existing warehouse receiving operations.

## My Inventory

Display inventory physically received and assigned to the customer.

Use actual package and inventory records.

Show:

- Item
- Quantity
- Available
- Reserved
- Storage status
- Warehouse
- Authorized location information
- Request Dispatch action

Do not expose confidential internal warehouse layouts to customers unless explicitly permitted.

## Request Dispatch

Create a real shipping/dispatch request from stored inventory.

Required flow:

Select stored inventory → choose quantities → choose destination → shipping details → quote → submit.

Ensure available inventory cannot be dispatched twice.

## My Shipments

Use the existing shipment records and follow the shipment-table design reference.

Provide separate functional filters for shipment states.

## Track Shipment

Allow a customer to enter or select their authorized shipment tracking number.

Show the actual latest status, route milestones, ETA where supported and proof of delivery when available.

Do not simulate GPS.

### Payments & Invoices

Use the Billing reference's styling but filter records to the authenticated customer's account.

Provide invoice viewing, downloads, payment status and genuine available payment workflows.

### Support, Notifications & Settings

Create functional pages using the same reference-based design conventions.

Support case creation, relevant notifications and authorized account settings.

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# 8. WAREHOUSE MANAGER DASHBOARD — COMPLETE RECONSTRUCTION

Use:

`TZLogistics Warehouse Operations Dashboard(2).png`

This is the Warehouse Dashboard's primary overview reference.

Do not replace it with the Admin Dashboard content.

Use the same overall visual language as Admin, but preserve warehouse-specific controls.

## Warehouse Overview

The top KPI row must follow the reference:

- Total Capacity
- Occupied
- Available
- Expected Arrivals
- Pending Receiving

- Awaiting Dispatch

Main content:

- Warehouse Occupancy
- Inventory by Category
- Storage Locations visualization
- Expected Arrivals
- Pending Receiving
- Picking Tasks
- Inventory Overview
- Dispatch Queue
- Recent Activity

Use a left sidebar with the reference's hierarchy.

# Warehouse Sidebar

## Dashboard

- Warehouse Dashboard

## Operations

- Storage Bookings
- Expected Arrivals
- Goods Receiving
- Inventory
- Picking Tasks
- Packing & Dispatch
- Warehouse → Shipping
- Stock Movements
- Locations & Bins

## Visual Warehouse Tools

Integrate the following without duplicating existing features:

- Warehouse Map
- Map Designer
- Racks & Bins
- Receiving & Putaway
- QR Labels & Scanner
- Locate Package

- Transfers & Picking
- Cycle Counts

### Management

- Warehouses
- Storage Rates
- Customers
- Employees

### Reports

- Occupancy Report
- Inventory Report
- Storage Revenue
- Activity Logs

Where a tool overlaps with an existing menu item, use an expanded sub-navigation or an integrated page, not two disconnected implementations.

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# 9. WAREHOUSE CHILD PAGES — FOLLOW EACH SCREENSHOT EXACTLY

These child screenshots are individually authoritative.

## A. Warehouse Map Designer

Reference:

`Warehouse Map Designer Dashboard(1).png`

Reconstruct:

- Horizontal drawing toolbar
- Interactive floor-plan canvas
- Dimension indicators
- Select, Pan, Wall, Door, Rack, Zone, Aisle and Shape tools
- Undo/Redo
- Grid toggle
- Zoom controls

- Save Draft
- Publish Map
- Right-side Selected Element inspector
- Layer Manager
- Floor selector and thumbnails
- Map legend

Use `react-konva`.

The canvas must be interactive.

Users must be able to place and configure real storage racks and zones.

Persist editable map geometry in Convex.

Do not use the screenshot as a static background pretending to be an editor.

# B. Racks & Bins

Reference:

`Warehouse Racks & Bins Dashboard(1).png`

Reconstruct:

**Top**

- Total Racks
- Total Bins
- Occupied Bins
- Available Bins
- Blocked Bins

**Left**

Rack list with occupancy indicators.

**Center**

Selected Rack view showing individual shelf levels and bins.

**Right**

- Bin Details
- Packages in This Bin
- Bin Capacity

- Quick Actions

**Bottom**

All Bins in Selected Rack table.

Support actual rack creation, editing and location selection.

A selected bin must correspond to a real Convex storage-location record.

# C. Receiving & Putaway

Reference:

`SWAHIVO Warehouse Receiving Dashboard(1).png`

Preserve the complete workflow:

1. Receive Packages
2. Print Labels
3. Assign Storage
4. Scan & Putaway
5. Complete

Keep the receiving form, generated-package list, QR scanner, floor map, shelf view and receiving table in their corresponding screenshot positions.

Connect receiving to actual bookings and shipment records.

A package must not become Stored merely because a location was selected.

Require authorized physical putaway confirmation.

# D. QR Labels & Scanner

Reference:

`Warehouse QR Labels Dashboard(1).png`

Recreate:

- Label Generator
- Scan Package
- Scan Location
- Scan History

- Print Queue
- Label Templates
- Settings

Maintain the screenshot's three primary sections:

- Label type and settings
- Package/location selection and print actions
- Label preview and scanner

Use:

- `qrcode`
- `@zxing/browser`
- `react-to-print`

Generate genuine QR data, not decorative placeholders.

QR scanning must retrieve the correct authenticated backend record.

The QR code should identify the canonical package or location rather than embedding private customer data.

# E. Locate Package

Reference:

`SWAHIVO Warehouse Package Locator Dashboard(1).png`

Recreate:

- Search toolbar
- Warehouse/floor selectors
- Top KPI cards
- Main interactive warehouse map
- Highlighted package location
- Package Details panel
- Rack R05 / shelf view equivalent
- Search Results table

When the user searches for a real package, load its last confirmed location from Convex.

Highlight the appropriate rack, shelf and bin.

Example:

`PKG-1042 → WH001-A-R05-L02-B03`

Treat this identifier as sample content only; do not hardcode it into production search results.

# F. Transfers & Picking

Reference:

`Warehouse Transfers & Picking Dashboard(1).png`

Recreate:

- Pending Picks
- In Picking
- Ready for Dispatch
- Internal Transfers
- Completed Today

Provide the two main workflow tabs:

- Picking (Outbound)
- Internal Transfer

Preserve:

- Select Pick List
- Pick Locations on Warehouse Map
- Current Pick Item
- Pick Actions
- Scan to Confirm Pick
- Pick List Items table

Use real package records, scan verification and authorized status transitions.

Do not let a package be picked twice.

# G. Cycle Counts

Reference:

`Warehouse Cycle Count Dashboard(1).png`

Recreate:

- Total Locations

- Counted
- Pending
- Discrepancies
- Active Count

Tabs:

- Overview
- Active Count
- Location List
- Discrepancies
- History

Main content:

- Warehouse Floor Map
- Location Status
- Progress
- Quick Filters
- QR Scanner
- Cycle Count Lists
- Count Summary
- Recent Discrepancies

Integrate real inventory verification.

Missing or mismatched packages must produce reviewable discrepancies rather than silently changing inventory records.

# H. Warehouse Reports

Reference:

`Swahivo Logistics Reports Dashboard(1).png`

Recreate:

- Warehouse Overview
- Inventory Report
- Location Utilization
- Movement Report
- Cycle Count Report
- Aging Report
- Custom Report

Include the same chart and table arrangement:

- Inventory Status
- Storage Capacity Utilization
- Package Movements
- Top Storage Locations
- Package Type Distribution
- Aging Report
- Recent Warehouse Activities

Use real Convex data and appropriate chart libraries.

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# 10. DRIVER DASHBOARD — COMPLETE RECONSTRUCTION

Use:

`TZLogistics Driver Dashboard(2).png`

This is the Driver dashboard's main visual reference.

Follow the common Admin-style visual system while maintaining a driving-focused workflow.

## Driver Overview

Preserve:

- Greeting and current assigned-work summary
- Start Navigation action
- My Assignments
- Scheduled Today
- In Transit
- Completed
- Issues
- Total Distance

Main content:

- Today's Route
- Route Stops

- Current Assignment
- My Assignments table
- Shipment Timeline

The geographic map is central to the Driver experience.

Do not replace it with a generic chart.

# Driver Sidebar

- Dashboard
- My Assignments
- Scheduled Pickups
- Active Shipments
- Deliveries
- Scan Package
- Delivery Confirmation
- Trip History
- Notifications
- Profile

# Build All Driver Child Pages

For pages without an individual supplied reference, derive their UI structure from the Driver Overview and corresponding Shipment, Transfer, and QR Scanner references.

## My Assignments

Use the existing assignments data.

Provide:

- Assigned pickups
- Assigned deliveries
- Scheduled date/time
- Customer and location
- Cargo details
- Current status
- Start/continue action
- Assignment detail page

## Scheduled Pickups

Display pickup tasks and scheduled windows.

Clicking an assignment opens a full detail page containing the customer address, cargo, verification instructions and permitted actions.

## Active Shipments

Show active trips assigned to the authenticated driver.

Display current status, stops, route milestones and location reporting availability.

## Deliveries

Show deliveries requiring action.

Allow authorized arrival, handoff and exception reporting.

## Scan Package

Reuse the actual QR scanner implementation.

A scan must identify a real package assigned to the applicable driver's task.

## Delivery Confirmation

Provide electronic proof of delivery:

- Recipient name
- Delivery timestamp
- Photo evidence where allowed
- Signature capture
- Notes
- Final confirmation

Preserve actual delivery records and audit events.

## Trip History

Display completed trips with actual distances when available, completion dates, shipment references and documented exceptions.

## Notifications and Profile

Provide working role-appropriate screens.

## Mobile Requirements

The Driver dashboard must be highly usable on mobile.

Keep:

- Clear navigation
- Large enough touch targets
- Readable assignment details
- Accessible scanner
- Visible delivery action controls
- Network-error handling
- Accurate location permission status

Do not show invented vehicle movement, trip distance or GPS accuracy.

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# 11. SHIPMENTS — IMPLEMENT SHARED REFERENCE

Reference:

`SWAHIVO Logistics Shipment Dashboard(1).png`

Implement the supplied design with:

- Shipment KPI cards
- All Shipments status tabs
- Search/filter controls
- Shipment table
- Shipment Details panel
- Shipment Timeline
- Tracking Map
- Items
- Cost & Payment
- Related Shipments
- Quick Actions

Status tabs must query the corresponding records.

For example, the Delivered tab must show only shipments with the appropriate actual delivered status.

Do not show the same shipments for Pending Pickup, In Transit, Out for Delivery and Delivered.

## Create Shipment

Reference:

`SWAHIVO Shipment Creation Dashboard(1).png`

Reconstruct the five-step workflow:

1. Shipment Details
2. Pickup & Delivery
3. Packages & Items
4. Services & Payment
5. Review & Submit

Use a full-page form.

Maintain the screenshot's multi-column form and right-side summary.

Support:

- Standard Shipping
- Storage + Shipping

Both service types must use the same underlying shipment infrastructure where appropriate.

**Standard Shipping** must not require a warehouse storage booking.

**Storage + Shipping** must link to real warehouse booking and inventory workflows.

Use actual customers, pickup/delivery locations, package details, service pricing and available carriers.

Do not preselect a fictional driver or vehicle.

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# 12. CUSTOMER CRM, TRANSPORTERS, BILLING & REPORTS

These are shared enterprise modules.

Reconstruct them using their individual reference images, then enforce role-based data access.

# Customer Management

Use the provided Customers CRM design where available.

Include:

- Customer summary KPIs
- Customer type/status filters
- Customer table
- Selected Customer Details
- Customer shipments
- Storage
- Billing
- Notes
- Customer analytics

Use real saved customer records.

# Transporter Management

Reference:

`Transporter Management Dashboard(1).png`

Implement:

- Total Transporters
- Active Transporters
- Total Drivers
- Total Vehicles
- On Deliveries
- Transporter list
- Region filters
- Coverage map
- Transporter Details
- Vehicles
- Drivers
- Assignments
- Documents
- Performance

Respect provider ownership and organization access.

# Billing

Reference:

`SWAHIVO Billing Dashboard Overview(1).png`

Recreate:

- Total Invoiced
- Paid
- Pending Payment
- Overdue
- Total Invoices
- Revenue Overview
- Revenue by Service
- Invoice List
- Invoice Preview
- Payment Information

Tabs:

- Invoices
- Payments
- Quotes
- Credit Notes
- Price List
- Usage & Charges
- Settings

Support authorized invoice creation, actual payment records and correctly calculated statuses.

Use real finance entries. Do not mark payments successful before actual verification or authorized manual reconciliation.

# Shared Analytics

Follow the Warehouse Reports reference for visual style where there is no more specific screenshot.

Allow each role to see only appropriate information.

Customer financial reports must not expose other customers.

Warehouse managers see operational warehouse reports.

Administrators see the broader organization reports they are authorized to access.

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# 13. PRESERVE THE PUBLIC LANDING PAGE

Reference:

`TZLogistics Warehouse & Shipping Platform(2).png`

This is a deliberate exception to the dashboard styling.

Maintain its separate public website composition.

Preserve:

- Main navigation
- Hero area with logistics imagery
- Warehouse + Shipping service cards
- Shipping Only service card
- Shipment tracking search
- Book Warehouse Storage
- Request Shipping
- How It Works section
- Public business information

Keep the established Swahivo branding and approved marketing copy.

Do not replace the marketing landing page with the Admin sidebar layout.

Connect public CTAs to the correct real application workflows.

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# 14. CROSS-DASHBOARD BUSINESS INTEGRATION

All four dashboards must share real backend records.

Do not create isolated, unrelated applications for Admin, Customer, Warehouse and Driver.

## Workflow A — Customer Books Storage

1. Customer opens Warehouse Bookings.

2. Customer selects a warehouse and submits a real booking.
3. Booking becomes visible to authorized Admin and Warehouse users.
4. Warehouse Manager reviews the booking.
5. Warehouse staff receive the physical packages.
6. Inventory records are created or linked correctly.
7. Staff assign verified rack/bin storage locations.
8. Customer's My Inventory updates from the same records.

## Workflow B — Warehouse Putaway

1. Warehouse staff identify a received package.
2. Print its genuine QR label.
3. Choose an eligible storage location.
4. Scan package and bin codes.
5. Confirm physical placement.
6. Update actual storage occupancy.
7. Make the package searchable on the map.
8. Update authorized inventory views and audit history.

## Workflow C — Customer Requests Dispatch

1. Customer selects stored inventory.
2. Customer creates a dispatch request.
3. Warehouse receives a picking task.
4. Staff locate packages using the map.
5. Staff verify picking.
6. Packages move to the correct dispatch state.
7. Existing shipment management receives the handoff.
8. A driver is assigned by authorized operations staff.

## Workflow D — Driver Delivery

1. Dispatcher assigns a legitimate shipment.
2. Driver sees it in My Assignments.
3. Driver performs permitted pickup confirmation.
4. Shipment status updates for Admin and Customer.
5. Driver progresses through authorized delivery milestones.
6. Driver records proof of delivery.
7. Shipment appears in Delivered views.
8. Customer and Admin see the final delivery status.
9. Finance receives the appropriate billable milestone.

## Workflow E — Billing

1. Storage or shipping generates valid billable charges.
2. Authorized invoice creation uses actual business and customer data.
3. The invoice appears in the correct customer account.
4. A real payment or approved manual payment verification updates its status.
5. Admin financial reporting updates.
6. Applicable business restrictions are enforced according to actual payment and contract rules.

**All five workflows must reference shared Convex data.**

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# 15. ROLE-BASED ACCESS CONTROL

Preserve existing authentication and membership models.

Apply secure backend permissions.

| Feature | Admin | Customer | Warehouse | Driver |
|---|---|---|---|---|
| Organization overview | Authorized | Own account | Assigned warehouse | Assigned work |
| Create shipment | Authorized | Own account | If permitted | No |
| Manage storage booking | Authorized | Own booking | Operational processing | No |
| Draw warehouse map | Authorized | No | Manager only | No |
| Receive packages | Authorized | No | Authorized staff | No |
| Putaway / Move inventory | Authorized | No | Authorized staff | No |
| Assign driver | Authorized | No | If delegated | No |

| | | | | |
|---|---|---|---|---|
| Confirm assigned pickup | Oversight | No | Warehouse handoff | Assigned driver |
| Confirm delivery | Oversight | No | Handoff only | Assigned driver |
| View invoices | Authorized | Own invoices | Permitted operational scope | Only if explicitly allowed |
| Manage pricing/settings | Authorized | No | Delegated only | No |

Enforce all sensitive read and write restrictions inside Convex functions.

Frontend hiding is not sufficient authorization.

Do not use mock admin logins or hardcoded preset roles.

---

# 16. REAL BACKEND AND DATA REQUIREMENTS

All operational UI components must connect to real Convex queries and mutations.

This includes:

- KPIs
- Charts
- Tables
- Search
- Filters
- Notifications
- Booking data
- Shipment data
- Inventory
- Driver assignments
- Storage locations
- QR scan results
- Invoice totals
- Revenue analytics
- Payment statuses
- Activity timelines

When no records exist, display a proper empty state.

Do not populate production dashboards with fabricated shipment totals or example customer information.

Use loading skeletons while data is loading.

Show genuine errors when backend operations fail.

Do not silently report successful mutations when Convex rejects them.

All date, currency, weight, status and quantity values must be formatted consistently.

---

# 17. VISUAL FIDELITY ENFORCEMENT

You must perform a screenshot-based visual comparison.

For each reference page:

1. Identify the matching implemented route.
2. Render the page at the same desktop viewport dimensions as the reference, approximately 1536 × 1024.
3. Capture a screenshot of the implemented page.
4. Compare it to the supplied reference.
5. Inspect differences in:
    - Sidebar dimensions
    - Header dimensions
    - Page margins
    - Card positions
    - Grid structure
    - Chart sizing
    - Table widths
    - Row heights
    - Fonts
    - Color usage
    - Status badges
    - Buttons
    - Maps
    - Panel alignment
6. Correct visible mismatches.
7. Repeat until major layout differences are resolved.

Use Playwright screenshots if available.

Use screenshot comparison tooling where practical. Pixel-difference tools are useful, but dynamic maps, fonts and real data can create irrelevant differences; prioritize structural and visual accuracy.

Do not claim pixel-perfect matching without conducting the comparison.

## Visual Acceptance Rules

A page fails visual review when:

- A major reference section is absent.
- The sidebar layout differs substantially.
- The page hierarchy has been changed.
- A table is replaced by cards without permission.
- A permanent reference panel is moved into a modal.
- Charts are omitted.
- The map is replaced by a decorative image.
- Important controls are missing.
- The layout uses unrelated styling.
- A child page is replaced by the dashboard overview.
- The implementation ignores its assigned reference screenshot.

Real production data can change text and chart content, but must not arbitrarily alter the visual composition.

---

# 18. RESPONSIVE DESIGN

The screenshots primarily define the desktop experience.

Desktop is the authoritative layout.

For tablet and mobile, adapt the same design rather than inventing a new design language.

## Desktop

Preserve the screenshot's full sidebar and content grid.

## Tablet

Use responsive columns and a collapsible sidebar.

Keep the primary content and operation controls accessible.

## Mobile

Use:

- Collapsible navigation
- Single-column sections where appropriate
- Touch-friendly forms
- Scrollable data tables when necessary
- Full-page task views
- Accessible scanner interface
- Clear primary actions

Do not hide critical operational actions just to simplify the mobile screen.

The Driver portal and Warehouse scanning workflows must be especially mobile-friendly.

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# 19. MANDATORY IMPLEMENTATION ORDER

Do not attempt to generate every page with generic templates in one pass.

Work in controlled stages.

### Phase 0 — Audit

Inspect the existing codebase, routes, backend and all supplied screenshots.

Produce the actual page-to-reference mapping and identify missing functionality.

### Phase 1 — Shared Design System

Align the existing reusable components with the Admin master reference.

Implement or refine:

- Dashboard shell
- Sidebar

- Header
- KPI cards
- Data tables
- Status badges
- Chart containers
- Search/filter controls
- Buttons
- Tabs
- Detail panels
- Empty/loading/error states

Do not break existing working pages.

## Phase 2 — Admin Dashboard

Reconstruct the Admin Overview and required Admin child pages using the respective screenshots.

Verify the interface.

## Phase 3 — Customer Dashboard

Reconstruct the Customer Overview and all customer child pages.

Verify booking, inventory, shipping, tracking and billing workflows.

## Phase 4 — Warehouse Dashboard

Reconstruct the Warehouse Overview and all reference-based warehouse child pages.

Implement functional maps, locations, QR labels, receiving, putaway, picking and reporting.

## Phase 5 — Driver Dashboard

Reconstruct the Driver Overview and all driver child pages.

Verify assignment, pickup, delivery and proof-of-delivery workflows.

## Phase 6 — Cross-Role Integration

Verify that actions performed by one role correctly update the records visible to other authorized roles.

## Phase 7 — Responsive and Visual QA

Compare implemented screenshots to the uploaded references, test desktop/mobile and repair layout discrepancies.

### Phase 8 — Production Readiness

Run authentication, authorization, operational, regression, and deployment-readiness checks.

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# 20. MANDATORY TESTING

At minimum, verify:

- Admin can navigate to every authorized child page.
- Customer can complete storage booking and shipment workflows.
- Warehouse receives the resulting operational records.
- Warehouse managers can draw and save real maps.
- Staff can assign a package to an actual storage location.
- Genuine QR generation and decoding work.
- Package search highlights the correct stored location.
- Authorized drivers receive assigned work.
- Driver pickup and delivery updates propagate correctly.
- Billing is connected to legitimate operational records.
- Filters and tabs return different appropriate datasets.
- Refreshing pages preserves real backend data.
- Unauthorized access to other organizations is rejected.
- The existing landing page is not damaged.
- No major page depends on demo data.
- No primary workflow has been replaced by popup cards.
- Implemented screenshots structurally match their corresponding reference images.

Perform suitable unit, integration and end-to-end tests.

Report tests actually executed and distinguish passing, failing, blocked and not-run cases.

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# 21. FINAL IMPLEMENTATION REPORT

When finished, provide a structured report:

| Field | Required information |
|---|---|
| Dashboard | Admin / Customer / Warehouse / Driver |
| Page | Actual implemented page |
| Reference | Exact uploaded screenshot |
| Route | Existing or implemented route |
| UI Fidelity | Pass / Issues / Not Reviewed |
| Frontend Functionality | Working / Partial / Blocked |
| Convex Integration | Verified / Not Verified |
| Authorization | Tested / Not Tested |
| Responsive Design | Tested / Not Tested |
| Remaining Issues | Specific details |

Do not mark an item complete solely because a component renders.

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# 22. ABSOLUTE FINAL DIRECTIVE

**THE ATTACHED SCREENSHOTS ARE THE DESIGN AUTHORITY.**

Your job is to reproduce and implement them.

The Admin Overview screenshot defines the main dashboard design conventions.

The Customer, Warehouse and Driver screenshots define their respective overview layouts and role-specific functionality.

The detailed Swahivo child-page screenshots define the exact page structures for shipments, receiving, racks, QR labels, package location, transfers, cycle counts, reports, transporters and billing.

Where a screenshot is available, follow it closely.

Where a screenshot is missing, derive the page from the closest existing reference without changing the established structure or design language.

Use dedicated full pages and permanent inline sections, not floating popup-card workflows.

Preserve the existing Vite, React, TypeScript, Tailwind CSS, shadcn/ui and Convex architecture.

Connect every applicable feature to real backend functionality.

Do not invent placeholder data as real operational records.

Do not use fake QR codes or simulated tracking.

Do not simplify or discard business workflows.

**RECONSTRUCT THE UI. PRESERVE THE ARCHITECTURE. CONNECT THE FEATURES. VERIFY THE RESULTS.**

Begin with a complete existing-project and screenshot-reference audit, then implement each dashboard and its child pages according to the instructions above.